Original article

Towards a history of the hammock: An Indigenous technology in the Atlantic world

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Abstract When Europeans arrived in the Western Hemisphere beginning in the fifteenth century, they learned that Indigenous groups across the Caribbean and South America valued few technologies as much as the hammock. Despite this, the technology has been neglected by scholars, revealing the shortcomings of frameworks that focus on 'commodities,' 'resources,' and even 'objects,' and underscoring the importance of centering subaltern technologies in Atlantic and European history. Much more than a bed, the hammock figured in healing, shamanic, and inter-social practices, including hospitality. It was precisely these intersubjective functions that enabled its transcultural movement. In settings of Indigenous healing and hospitality, early modern Europeans learned to value hammocks, and they swiftly incorporated them into the colonial apparatus, often in ways that furthered the dispossession of the communities from which the technology came. Yet even as the hammock became a settler-colonial technology, it still bore traces of Indigenous uses and meanings.

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Introduction

In the sixteenth century, Jean de Léry, a French Protestant missionary who lived among the Brazilian Tupinamba for two years, learned to love an Indigenous technology: the hammock. In his *Histoire d'un voyage faict en la terre du Brésil*, he asked if 'those who have tried them out whether you don't sleep better on these cotton beds than on our ordinary ones, especially in summer' (1990, 160). Léry's rhetorical question implies the widespread 'trying out' of the hammock by Europeans and the pleasure and benefit it brought these early adopters. Though the hammock was marvelously novel to Europeans like Léry, it had an ancient American pedigree, dating back at least 4000 years from the present, and its use was broadly distributed across lowland South America. Europeans, after exposure to hammocks in rites of Indigenous hospitality, quickly learned their value. The technology swiftly became an essential part of settler-colonial culture, and before long, spread to other parts of the Americas, Europe, Asia, and Africa. Today, it is used across the globe.

The global historical importance of the hammock is not reflected in the relatively meager attention it has received from scholars, with some important exceptions (Mira Caballos 1997; Niño Vargas 2014; Matthews Cascon and Fernandes Caromano 2020; Zimmerman 2017; Samson et al. 2023; Wiedemann 1979). This neglect is related to a 'stagist' view of technological history that continues to structure and pervade the historiography of the colonial Americas. In this view, Europeans had attained a more advanced stage of cultural development than Indigenous groups. Most historians writing today would not assert, as did Alfred Crosby, that 'when Columbus arrived, even the most advanced Indians were barely out of the Stone Age' (2003, 21). But the effects of this paradigm still linger in the scholarly tendency to refer to Indigenous technologies as 'resources' or 'biota,' a framing that characterizes these objects as emerging fully formed from the natural world, rather than from applied human knowledge. Histories of 'commodities,' too, which typically begin at the point in which Indigenous technologies have economic importance for Europeans, tend to elide Indigenous contributions (Norton 2017). Moreover, even ethnohistorical scholarship focused fully on Indigenous communities has not freed itself from an evolutionary model that marginalizes some Indigenous technologies in favor of others. In this scholarly tradition, certain civilizations and their technologies—such as the monumental architecture and writing systems of Mesoamerica and the Andes—are marshaled as proof of their evolutionary advancement and 'civilization,' contrasted explicitly or implicitly with allegedly less advanced cultures of the circum-Caribbean and lowland South America (Norton 2024).

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We advocate a history of technology that does not privilege recent inventions, hard materials, and mechanization; instead, we define *technology* as simply the intentional transformation of matter.[1] This reframing has several advantages. First, it enables us to uncover influential forms of Indigenous ingenuity. The hammock—so widespread in colonial sources as almost to be beneath scholarly notice—is a prime example of the kind of Indigenous object that has been occluded in histories of technology, despite its demonstrable importance. Recognizing the ubiquity and importance of the hammock, in early modernity and today, in turn challenges a view of globalization as a unidirectional relationship in which Europe dispersed more 'advanced' technologies to a more 'primitive' Indigenous America. The second advantage of our more expansive definition of *technology* is that it brings into focus a domain of human life often excluded from histories of technology: care work, frequently done by women and so quotidian as to escape attention. Indeed, the limited amount of recent scholarly work on hammocks has suggested examining them, like other 'mundane and material' technologies of care, might allow us to see 'minoritarian power' in new ways (Samson et al. 2023, 826).

1 Here, we further develop arguments about technology outlined in Norton (2017), building on Gell (1988); Bray (2008).

It is hard to overstate the hammock's centrality to the Indigenous cultures of lowland South America and the Caribbean, and here we trace its use across the entire region and several centuries. This very broad array of sources risks flattening out regional/temporal texture in hammock use, which had many variations over an extensive space (first restricted to lowland South America and Circum-Caribbean before diffusing to other parts of the Americas, Africa, Asia, and Europe) and over the longue durée. However, this risk is counterbalanced by the benefits of this approach. First, because little scholarship exists on the hammock, a broad approach is needed to map general contours of its meanings and uses. Second, this approach affords methodological benefits in our attempt to recover this technology's early modern Indigenous history. Because early colonial sources were primarily produced by European outsiders, who inevitably described Indigenous societies through their own cultural filters, a full picture of the importance of hammocks in Indigenous communities is not available in a single source. Yet, when examined in aggregate, a fuller picture does come into view. We supplement these sources with later ethnographic accounts, and, above all, a twenty-first-century Indigenous work by Yanomami shaman Davi Kopenawa (Kopenawa and Albert 2013). We do not use the later accounts to 'upstream,' or assume that more recent accounts can fill in lacunae in earlier sources. Rather, these later sources, particularly Kopenawa's memoir, help us see and interpret details in early modern sources that might have otherwise gone unnoticed. Moreover, Kopenawa's reflections on spirituality and care that so often

center the hammock also serve as a reminder that hammocks' centrality in Native and, eventually, settler culture is only a 'discovery' for non-Indigenous outsiders.

The hammock enabled a broad array of cultural functions related to care, conviviality, and intimacy (Walker 2009; Samson et al. 2023; Norton 2023). The hammock's key material feature was its membranous nature: the way it marked off a space as personal and autonomous, while simultaneously allowing proximity to and connection with others. It acted as a second skin or a fabric pellicle, one that responded to the body's pressure: solid enough to protect the body, while porous enough to allow the passage of sound, sweat, warming heat, and/or cooling air. These material aspects of the hammock cannot be separated from its cultural functions and the ontologies reflected in and created by them. After discussing its Indigenous uses and meanings, we turn to another key chapter in the hammock's history: its early appropriation and usage by settler colonists. Hammocks were central to alliance building in Native communities; these practices were extended to early European colonists, who almost immediately became socialized into them. Hammocks were also at the core of the native healing practices offered to many early outsiders. In other words, it was precisely the technology's intersubjective functions—its role in hospitality, alliance-building, and care for the sick—that enabled its transcultural movement. In these settings, Europeans quickly apprehended the benefits of the hammock, and in a sad irony of history, we show how the hammock was swiftly incorporated into the colonial apparatus, often in ways that furthered the dispossession of the very communities in which the technology had originated.[2] We show, however, that even as the hammock became a staple technology of settler-colonial America, it never became fully deracinated, and it still bore traces of its original Indigenous uses and meanings.

2 For an archaeological perspective on this process, see Samson et al. (2023).

The hammock in early modern Indigenous America

Hammocks, like staple foods such as manioc and psychotropics such as tobacco, were and are one of the most-widespread Indigenous technologies in lowland South America and the Caribbean. In aggregate, twentieth-century ethnographic reports on Brazilian Indigenous groups show that sixty-seven of the eighty-eight groups represented in the documentation used hammocks (Wiedemann 1979). Early modern sources corroborate this picture of the hammock's widespread use. Outsiders traveling to Indigenous areas from the fifteenth century onward commented on their use among groups throughout the western and eastern Caribbean, the Atlantic coast, throughout the Orinoco watershed, and in the Amazon



Fig. 1: Cotton hammock, late seventeenth century, Brazil. Amsterdam, Rijksmuseum, NG-NM-4982. Image credit: Rijksmuseum, Amsterdam

basin and its tributaries.[3] A recent archaeological dig also suggests their great antiquity: in northern Brazil, a hammock comprised part of a Late Middle Holocene burial assemblage, dating from between 3750 and 8590 years ago (Solari et al. 2022). Across its spread, there were different names for the technology, *hamaka* in Arawak, *lelétêbou* in Kalinago (Breton 1665, 335), and *inis* in Tupi (Léry 1990, 160), among others: in this article, we use the now-common term *hammock*, derived from the Arawak.

Though the hammock was widespread, it was not invariant. Few hammocks from the early colonial period survive—this is perhaps not surprising, given the low survival rate for textile objects in the tropics—but a small number of hammocks were preserved in early modern European *wunderkammern*. They include examples at the Rijksmuseum (Amsterdam) (Fig. 1), the Ashmolean (Oxford), Skokloster Castle (Håbo Municipality), and two in the National Museum of Denmark (Copenhagen).[4] These can give us a sense of the range of hammock sizes, constituent materials, and appearances: they range in size from 7.5 feet to 17 feet; they are made of cotton (Rijksmuseum and Ashmolean) and palm fiber (Skokloster); and their styles can be roughly divided into solidly woven (Rijksmuseum) and more net-like (Skokloster).

The division in hammock structure suggested by the surviving examples was also noted by early modern observers and more recent visitors alike

3 Western Caribbean: Las Casas (1875); Oviedo (1535, fols. 47v–48r). Eastern Caribbean: Breton (1665); Tertre (1654). Tierra Firme: Oviedo (1959, 42); Kraemer (1996). Tupi-dominated areas: Thevet (1558); Léry (1990). Orinoco watershed: Cei (1995); Ruiz Blanco (1892); Gumilla (1745); Gilij (1965). Western Amazon: Figueroa (1904); Eder (1888); Albis (1991).

4 Rijksmuseum example: Object number NG-NM-4982; Ashmolean: AN1685.B.422; Skokloster Castle: 7061_SKO; National Museum of Denmark: EHc35 and EHc36.

(Las Casas 1875, 3:310 and 5:521; Bechamel and Grillet 1698, 55; Wiedemann 1979). Other early modern accounts shed more light on these two forms. The French Jesuits Jean Grillet and François Béchamel provided a detailed account of the manufacture of solid-weave cotton hammocks in Guiana in 1674:

> They weave 'em after the following Manner . . . All their Implements are two round Sticks, about 8 or 9 Foot long, and 3 or 4 Inches Diameter; the two Ends of one of these round Sticks lie a-cross upon two Pieces of Wood, about 8 or 9 Foot from the Ground, more or less, according to the Length the Work-woman is order'd, or designs herself to make her Bed. The other round Stick hangs directly under this, and 'tis to these two round Sticks that the Warp of the Hamock is fasten'd, after which they have a kind of Shuttle, which they put through the Threads, to weave it after the Manner of our Cloth (1698, 56–57).

This description provides rough background for the Brazilian hammock in the Rijkmuseum, which is made of cotton and capped on either end with fringe and beautiful detail. Grillet observed women making hammocks; this gender pattern is corroborated by other early modern sources as well.[5] The second form of hammock, more like a net, is described in a British Royal Society inventory.

> A Hammock: Like a Great Net, with several small tassels on the sides, and two huge ones at the ends. Between which, 'tis fifteen feet long. The Weft, seven feet; and about as broad. It consisteth of twisted Thread, as thick as small Pack-thred; made of the Barque of the Coco-Tree, and of the Rind of the Nut. Not Netted; nor Woven with Warp and Woofe; but after the manner of Bobbin-Work. At both ends, the Weft gather'd up into several small Ropes, and those at last into greater. (Grew 1681, 372)

Though not a first-hand account of Indigenous manufacturing, this description may provide background for understanding the structure of 'net-like' hammocks, like the surviving example at Skokloster.

The hammock cannot be treated in isolation; rather, like all technologies, it is modular (Norton 2017, 26). The hammock might be best conceptualized in radial terms, as a hub from which emanates a host of different technological spokes. These spokes include: agroforestry, the knowledge and methods for managing and benefiting from many wild and cultivated species; fiber-working, the knowledge, techniques, and instruments (such as spindles) needed to transform plant species into usable threads and cords; dyeing, the knowledge and techniques used to color fibers or decorate finished hammocks; weaving, the techniques and

5 For Brazil, Guyana, and Dominica, see Léry (1990, 160); Raleigh (2006, 200); and Breton (1665, 192, 201).

infrastructure (such as looms) used to create hammocks; cleaning, the techniques and knowledge used to process and transform plant material into cleaning agents; and architecture, the house designs that supported hammock use.

Chief among these connected technologies was agroforestry. First and foremost, hammocks were woven from threads made from plants, typically varieties of cotton and palm, which had been used in South America for millennia as both foraged and domesticated plants (Piperno 2011). Inga Wiedemann, in her twentieth-century survey of Indigenous hammock-making techniques in Brazil, identified nine different plant species used to make hammocks, including agave and at least four different species of palm, with cotton and palm fibers generally predominating. She surveyed sixty-five groups: twenty-four used cotton fibers, sixteen used palm fibers, and twenty-one used both fibers. Other fibers were also frequently mixed in (Wiedemann 1979, 113–24). Europeans admired the quality of the fibers Indigenous people made from cotton, often comparing them to silk. In his abbreviated 1526 natural history, often referred to as the *Sumario*, Gonzalo Fernández de Oviedo, infamous conquistador and hammock acolyte, described hammocks that he saw and used among Indigenous groups on Tierra Firme (coastal Colombia and Venezuela) in the early sixteenth century: 'The beds in which they sleep are called *hamacas*, which are pieces of well-woven cotton cloth and of good and pretty tapestry. Some of them are very thin' (Oviedo 1959, 42). Léry wrote that he 'brought to France some of the finer cord, so well spun and tightly twisted by these same women that when I had a white cloth doublet stitched with it, everyone who saw it thought it was fine purled silk' (1990, 160). In many places, the connection between cotton and hammocks was so tight as almost to be synonymous. Raymond Breton, a French missionary who lived among the Kalinago between 1642 and 1654, recorded an extensive vocabulary connecting hammocks to cotton (1665, 325–26). Francisco Javier Eder, a Jesuit missionary who lived among the Moxos in the Bolivian Amazon in the mid-eighteenth century, described palm fiber being used to make hammocks (1888, 44).

Though cotton and palm fibers seem to have been the primary materials used to make the body of the hammock, they were also supplemented by a range of other plants used in other aspects of the production and maintenance of hammocks. For instance, agave and hibiscus plants sometimes provided the material to make the strong support cords for hanging hammocks. Oviedo wrote in 1526 that the hammocks in Tierra Firme 'are covered with long cords made of *cabuya* [century plant] and of *henequén* [sisal hemp]' (1959, 42). These latter cords, he explained in his longer natural history of 1535, were 'long and detachable and run along to the ends of the hammock from a knot . . . made like the ends of a

bowstring, and like this they mount it, hanging it from the hammock's end' (1535, fol. 47v). In Hispaniola, according to Oviedo, these attachment cords were made of bark from a tree called *damahagua*, the latter stronger 'than henequen or hemp' (1535, fol. 47v).[6] Sixteenth-century observers noted the same or similar bark on the continent (Aguado 1906, 440; see also Wiedemann 1979, 107). Other plant materials were used to clean hammocks. Léry wrote that when the hammocks became soiled, the women cleaned them with a wild fruit 'in a form like a flat pumpkin, but much bigger,' out of which they made a soap so effective that 'the beds come out as white as snow or as fuller's sheets' (1990, 160).

6 Unless otherwise noted, translations are our own.

In addition to its links to agroforestry, the hammock also had a reciprocal relationship with architecture. Oviedo noted that 'if one sleeps in home, posts or supports of this *bohio* serve, in place of trees, to hang these *hamacas* or beds' (1535, fol. 47v). Léry, in Brazil, noted the intentionality of this connection; hammocks were attached 'to wooden beams set cross-wise in their houses expressly for that purpose' (1990, 160). Unlike European sleeping technologies, however, hammocks were highly portable. Many sources indicate that there was considerable flexibility in hammock placement within, between, and outside of Indigenous dwellings, often reflecting shifts in familial, hospitable, or sexual relationships. Léry observed that the hammock's portability allowed the Tupi to more comfortably conduct activities away from home: when 'they go to war, or sleep in the woods during a hunt, or by the seashore or by their fishing streams' they 'hang them between two trees' (1990, 160). In the Caribbean, the Huguenot minister Charles de Rochefort observed the construction of temporary shelters (*aioupa*) during voyages, erected in a 'half hour,' with simple leaf roofs mounted on 'forks' driven into the ground; hammocks could be attached to these supports (1666, 320). These examples show us how Indigenous architectures, both permanent and temporary, were related to the hammock's portability. The textile shell could be incorporated into domestic architecture, but it did not have an exclusive or permanent relationship to any given space and could easily be removed and attached to supports elsewhere.

Though hammocks could vary in their materials and construction, the design's core was invariant: a textile sling that fit loosely around the body of its occupant. This quality—its solid yet porous wraparound membrane—facilitated many of the hammock's cultural functions and meanings in Indigenous life. This membranous quality can be seen when attending to the role that hammocks played in the birth process. Evidence from Breton's Kalinago-French dictionary reveals that the word for hammock was cognate to the word for womb: *iále* might be translated as 'womb'; *liále tomáli ácae* is 'the belly of an earth vessel' (which, not incidentally, could also be used as mortuary vessels); *ialicátobou* is 'the

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place of birth, it is also the pellicle that envelopes the child in the stomach of his mother'; *iále ibaou:ánale* is 'is the word for what is done by [he] who receives a countryman [*paysan*], such that he presents a bed of cotton'—in other words, a hammock (Breton 1665, 266–67).

The hammock, here, is linguistically analogous to an embryonic sack: a divider solid enough to separate the child from its mother, but porous enough to allow sustenance to flow across the membrane from one being to another. One early modern source indicates that children moved directly from the womb into a baby hammock, a material practice that underlines the connection between the two organic pellicles (Rochefort 1666, 338). Ethnographer Harry Walker has discussed the role of baby hammocks in the birth process in an Urarina community in the Peruvian Amazon:

> A baby enters the world in a highly vulnerable and ambiguous state of existence, and the fabrication and use of the baby hammock form part of an extensive series of parental interventions intended to form and fortify the body, protect it from disease, and ensure its successful entry into full social personhood. The mother weaves the hammock from palm fibers just prior to leaving the purpose built annex (*jata*) of palm leaves in which she gave birth and must reside with her child in isolation until its umbilical cord falls (2009, 78).

Walker's analysis of modern practices echoes the philological traces found in Breton's dictionary: in both cases, the hammock is conceived as a kind of post-natal extension of the womb, a second membranous container that continues to protect the infant body.

Hammocks were also fundamental for a person's final transition, the passage from life to death. Archaeological, early modern, and contemporary sources alike attest to how they became a resting place before death and a burial shroud after (see Kopenawa and Albert 2013, 202–4; Farabee 1924, 81–82, 171–72; Roth 1915, 155). Indeed, the oldest known surviving hammock seems to have functioned as a burial shroud (Solari et al. 2022, 7–8). On Hispaniola, according to Columbus, hammocks sometimes figured in assisted death: elderly, ailing *caciques* were 'put in a hammock which is their net bed, and they put water and bread next to their heads, and they leave them alone, not seeing them again' (Arrom 1999, 45). Léry noted that 'worthy elders' were buried shrouded in their 'cotton beds,' a practice illustrated in printings of his text in Europe (1990, 175). In 1585, a Spanish official describing practices of Indigenous groups in the Caracas region wrote that they buried their dead in a hammock, putting 'a fire beneath it in order to dry out' the corpse (Latorre 1919, 80; see also Las Casas 1875, 521–22, 524; Albis 1991, 183). Breton included an expression (*nekeataíti*) that he translated as both 'he died in his bed

[hammock]' and 'the beautiful death' (1665, 371). His text also reveals the way the hammock after death mirrored the function of the embryonic sac before birth. He wrote that Kalinago were buried 'in a beautiful bed of cotton' put into a hole in the ground and the deceased are in a 'posture' that 'they had in the womb of their mother'(Breton 1665, 237–38, 371; see also Tertre 1654, 455; Rochefort 1666, 348). Once buried, the membranous hammock would eventually break down and decompose, allowing for the reintegration of the individual with undifferentiated, or less differentiated, organic matter. Given the hammock's role in birth and death, it is not surprising that it also featured prominently in other kinds of life-cycle transformations. These include puberty rituals, the *couvade* (a period of recuperation for fathers of newborns), and initiation rites for newly chosen leaders.[7]

The hammock facilitated transitions between life stages like birth, puberty, leadership, and death. But it also facilitated more quotidian shifts in the body: sleep, dreaming, entering hallucinogenic states, and healing. Most obviously, the hammock facilitated the shift between states of consciousness that we know as sleep, and many Indigenous words for the technology belong to this semantic field, including the Arawak word *hamaka* (Goeje 1928, 235). European observers were impressed with hammocks as a sleeping technology in several ways. First, they noted that hammocks prevented their occupants from getting chilly or overheated (Benzoni 2017, 29). They also noted their usual position near fires, which played a role in the second thing Europeans loved about hammocks: they kept away pests. Breton explained that 'the *Sauvages* make fire beneath them at night to defend against mosquitos and against coolness' and that the slings kept away 'snakes, ants, and other noxious critters [*bestioles nuisibles*]' (1665, 193, 233). Las Casas made a similar observation about their pest-repelling abilities, offering this as a hypothetical explanation for the absence of lice in the Americas (Las Casas 1875, 5:348). Some Indigenous groups even used them to protect themselves from jaguar assaults, by suspending them high in trees (Eder 1888, 112). Contemporary sleep researchers have borne out the observations of these early modern colonizers, showing that hammocks encourage better sleep than beds (Bayer et al. 2011).

Hammocks also facilitated a relaxed sociability that some Europeans characterized as sloth or laziness, as evidenced by Breton's dictionary. The missionary impugned what he considered to be an excess of 'leisure' among the Kalinago and connected this to their hammocks:

> They work at their leisure: they go in the morning after their breakfast, one to the coast, another on a hunt, but the weather needs to be good. When they prepare their field they go with their axes, and

7 On *couvade*, see Rochefort (1666, 337); Prudhomme (1798, 354); Roth (1915, 321). On initiation rites, see Carvajal ([1648] 1892, 335–36); Rochefort (1666, 315). On puberty, see Prudhomme (1798, 346); Roth (1915, 309–11).

> clear everything in a morning, but they drink the rest of the day and all of the following night. They will take a year to make a pirogue. While they rest in their hammocks, they make baskets, talk, play the flute, *voilà* these are their occupations. (Breton 1665, 201; see also Prudhomme 1798, 345)

In the eighteenth century, Jesuits similarly admonished Indigenous men in the Orinoco for 'lazing around' in their hammocks and excessive drinking while the women labored (Gilij 1965, 2:35). These comments, as well as those found in other outsider texts, demonstrate that the hammock could be a space for making baskets or arrows.[8] European observers, however, often characterized this work not as productive or industrious but as indolence, a sign of the low value placed on Indigenous technologies manufactured in hammocks. But we do not argue that hammocks were actually only spaces of 'productivity' of a sort that was illegible to Europeans; they seem to have often been genuine spaces for daytime rest and contemplation. Europeans, however, typically saw daytime rest in hostile terms. Colonial descriptions of leisure in hammocks are thus a revelatory index of a profound divergence among European and Native concepts and beliefs about labor, leisure, and pleasure. That hammocks were a site of private rest and contemplation among Indigenous peoples would eventually become the basis of European and settler-colonial judgment of associating hammocks with indolence, a motif that lingers in contemporary culture.

8 For arrow-making in hammocks see Léry (1990, 160), and later examples in Waterton (1879, 130, 368, 37).

Hammocks also facilitated another kind of bodily transition: entry into dream-states. In Davi Kopenawa's extensive account of his life and Yanomami philosophy and values, he describes the hammock as facilitating trance or 'ghost' states: 'I remained lying on my hammock for a long time, not moving. I became a ghost and when night fell I dreamt constantly' (Kopenawa and Albert 2013, 43). These dreams are not just a change in his own body; Kopenawa sees them as an opportunity for intersubjective contact between humans and spirits. In one passage, he describes his 'image' being seized by the '*xapiri*' and speeding 'far into the distance . . . onto the sky's back, where the ghosts live, or in the aõpatari ancestors' underground world' (2013, 42–43). Bruce Albert, Kopenawa's translator and editor, glosses this, saying that the Yanomami dream (*mari*) 'is considered a state of temporary wandering of the person's inner image/vital essence (*utupe*) outside of his body' (2013, 496n3).

The hammock is key to these states, which are often triggered by a hallucinogenic plant called *yãkoana*. Kopenawa describes the drug's effect as 'making me die' (Kopenawa and Albert 2013, 42) and from a reductive Western perspective, it might seem that the 'ghost state' is primarily caused by consuming the hallucinogenic plant (2013, 42). Yet,

Kopenawa's account suggests that the hammock, too, is a technology of transformation. The hammock is a key part of his descriptions of trance preparations: it puts the body into a receptive state and thus frees the mind to see and communicate with *xapiri* (2013, 85, 133). The hammock also preserves and supports the body while the 'image' or 'inner part' travels with the *xapiri*: 'your body remains prone in its hammock but the *xapiri* fly off with our image and show us unknown things. They carry our memory with them' (2013, 80). In another passage he similarly asserts that 'our real inner part is there, a great distance away from our skin, which is here, lying in our hammock!' (2013, 119). In all these descriptions of trances, the hammock appears as a technology that conditioned the body into a state receptive to spirits.

There is no early modern equivalent to Kopenawa's beautifully detailed account of shamanic trance states and *xapiri* visitations, but some sources make a connection between hammocks and dreaming or trancing. Oviedo described the way hammocks were a site for hallucinogenic trances, as well as for dreams (1535, fol. 47r). When describing preparations for important rituals involving tobacco, Breton mentioned that the participants retreated to their hammocks while fasting (1665, 203). European printers often used images to underline this connection between shamanic plant trances and hammocks. Benzoni's book, for example, included a woodcut of a shaman blowing tobacco smoke on a man prone in his hammock, and André Thevet's *France Antarctique* (1558) illustrates a shamanic ritual in one (Figs. 2 and 3).

The hammock was also a key element in healing practices, which were often entangled with shamanic rituals. A watercolor in the *Histoire naturelle* depicts a healer treating a man wounded by an arrow while he lies in a hammock (Fig. 4). A missionary described the 'manner of healing' in Guayupe communities in what is today Colombia in the sixteenth century: 'the patient' was put 'in a hammock in the air' with 'two fires with much kindling on either side,' and then the healer arrived 'and began to blow and speak certain superstitious words in his language and with this and with the fires burning . . . they have them there until [the sick person] dies or restores his health.'[9] One of the hammock's most important roles was facilitating movement between different kinds of mind-body states.

Connected to the way it enables intersubjective relationality between the dreaming body and spirit visitations, the hammock fostered intimacy of various kinds between people. Or, put slightly differently, individuals were both connected to and split from the social collective by the hammock's membrane. Paradigmatically, when a person was lying in their hammock in the company of others, the hammock—like the uterine sack in the womb—kept the individual autonomous while also allowing for co-

9 Aguado (1906, 443). See also Cei (1995, 78); Breton (1665, 261); Albis (1991, 197); Roth (1915, 182, 207, 321).

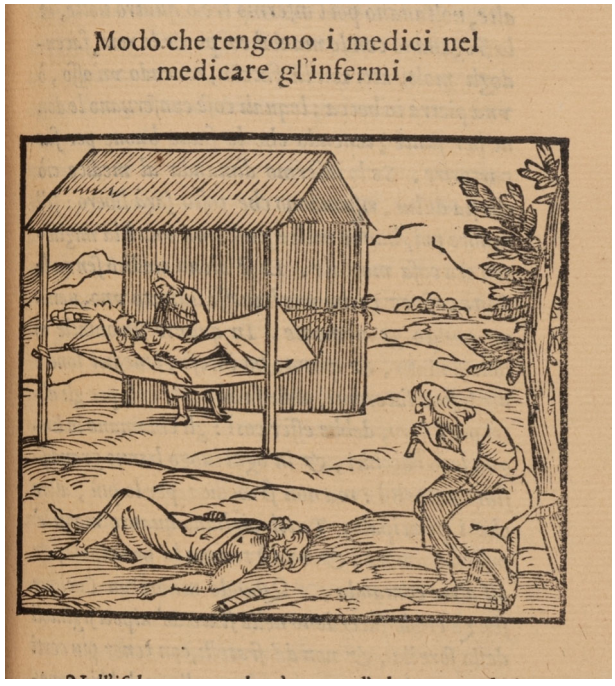


Fig. 2: Woodcut of shaman blowing tobacco on prone man in hammock, from Girolamo Benzoni's *La Historia del Mondo Nuovo* (Venice, 1565), 56. Image credit: Huntington Library, San Marino, CA

presence in the form of conversation and shared space. The closer others were, the greater the intimacy, with the greatest intimacy being that of the shared hammock. In Kopenawa's account and early modern sources alike, circles of intimacy form in roughly concentric circles around, and even inside, the hammock.

At the outermost level, alliances were articulated through gifts and trades of hammocks, reflected by their presence in gift exchanges between Indigenous and European groups, as discussed below. To foster deeper friendship, Indigenous hosts invited their guests to rest in hammocks. In Guyana, Raleigh observed a visiting Indigenous 'stranger who had been up the river in trade' being welcomed to lay in a hammock near a local man, where he was served *chicha*, an alcoholic beverage (2006, 126–27). Similar examples abound in Breton.[10] These hammock-centric, inter-Indigenous rituals of hospitality were extended to Europeans, in the hopes they would become friends, allies, and traders. Jean de Léry learned directly about the way the Tupinamba welcomed into a village 'a visitor' by bringing him to the house of a prominent man to be 'seated on a cotton bed suspended in the air' who then 'remains there for a short while without saying a word' (1990, 164; similar example in Farabee 1924, 69).

10 Breton (1665, 266). For other greeting rituals related to hammocks, see also pp. 24, 155, 214, 345.

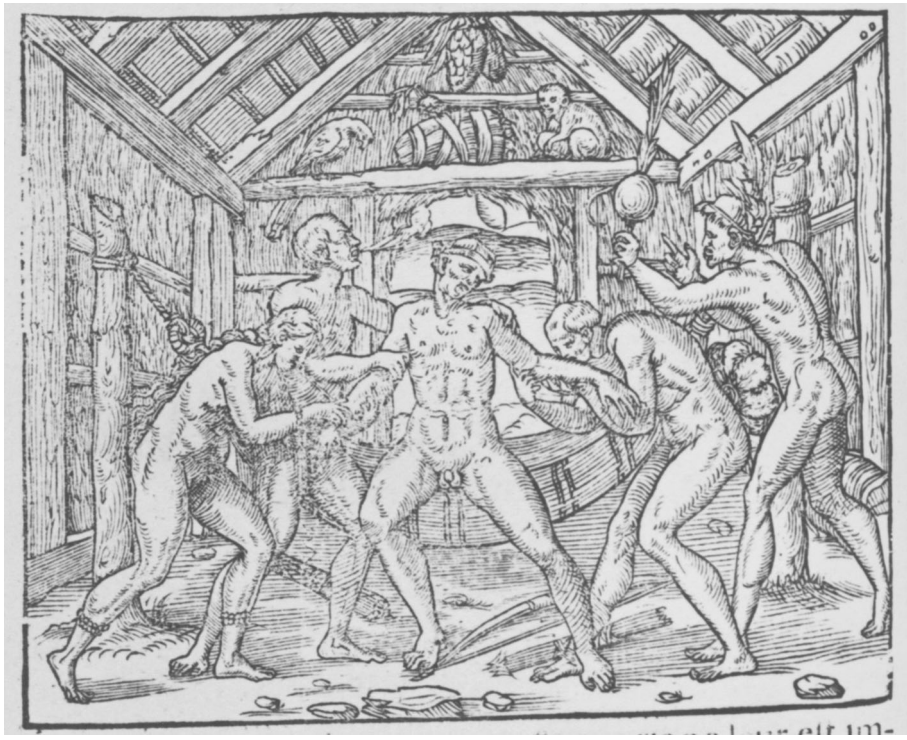


Fig. 3: Shamanic ritual taking place in a hammock. From André Thevet, *Les Singularitez de la France Antarctique* (Paris, 1588), 88–89. Image credit: Huntington Library, San Marino, CA

Similarly, Kopenawa recollects the way elders in his community were receptive and friendly to white settlers, offering an 'invitation . . . to tie their hammocks to the posts of our house' (Kopenawa and Albert 2013, 187). Conversely, when expressing dissatisfaction with a relationship and estrangement, Kopenawa describes being unwilling to have hammocks in proximity (2013, 334).

This hospitality extended to other-than-human beings. When a Kalinago shaman (*boyé*) made offerings to a spirit, he made sure to 'present him with a bed [hammock] so he could sit' along with 'the offering of food and drink' (Breton 1665, 216–17). Kopenawa, describing the intimacy between the *xapiri* and the thunder beings, also shows us how hammock proximity could be a sign of friendship or familial relations, writing that 'the spirits stretch out in their hammocks to demonstrate their friendship [to the thunder beings], as we do with a brother-in-law' (134).

Early modern accounts also offer glimpses of some of the inner circles of hammock relationality. Europeans marveled at how large numbers of people could sleep near one another in hammocks while still maintaining some privacy. In describing the 'communal house of the *Sauvages*' (the

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Fig. 4: Watercolor of an Indigenous man, wounded by an arrow, being tended to in a hammock, from *Histoire naturelle des Indes,* fol. 92r. Image credit: Morgan Library, New York, NY

sleeping quarters of adult men), Breton wrote that they could accommodate '100 or 120 beds of cotton where they rest peacefully with perfect intelligence without quarrels or noise' (1665, 477). Francisco de Figueroa, a Spanish missionary visiting the Maynas region (Peru) in the western Amazon in the late eighteenth century, was similarly impressed. He commented that a colleague 'counted in a single lodging 108 hammocks and after that I saw an even greater number . . . in other homes forty to sixty' (1904, 132). 'Everybody,' including 'the newborn, the wife, the husband and everyone,' he noted, 'slept in their own hammock with a little fire on the ground towards the feet' (1904, 132). Hammocks simultaneously facilitated a collective and private experience.

The most interior layer of intimacy was the act of sharing a hammock, reserved for sexual relationships between lovers and domestic partners, and close familial relationships, such as those between parents and children. The *Histoire naturelle*, for example, reveals hammocks'

importance for courtship rites in the Circum-Caribbean in the late sixteenth century.[11] The *Histoire* explains and illustrates how a young man behaves when he wants to marry a woman (Fig. 5). He approaches the house of the father of 'the daughter whom he loves,' taking with him 'all his equipment, namely his canoe or boat, bow, arrows, "chichorne," or fish-nets' (Kraemer 1996, 270 [fol. 113r]). After 'making his reverence to the father and the daughter, he says "Hai Hai," which means how are you?' and 'having done that, he leaves his whole equipment and utensils in the house and goes away to sleep in a hammock or bed and to rest until the next morning' (1996, 270 [fol. 113r]). The next day he hunts and gives the game 'to his beloved or sweetheart to please her and make her cook it' (1996, 270 [fol. 113r]). Kopenawa, too, expresses sexual longing in terms of the 'desire' for 'crossing the house on all fours to secretly join them in their hammocks' (Kopenawa and Albert 2013, 40). He also reminisces about the time when his future father-in-law sent the woman who he would marry 'to tie her hammock next to mine and we really became closer' (2013, 244).

Family members, too, shared hammocks. This of course, began with the hammocks that mothers created for their infants and used to carry them on their bodies. As they grew older and larger, children might share their parent's hammock. Kopenawa recalls an early childhood memory when 'the warmth of the fire eased me to sleep in my mother's hammock' (Kopenawa and Albert 2013, 40, 244, 34; see also 246, 331). A similar scene is evoked in the watercolor painted by a nineteenth-century missionary who lived among a Guaki community in the Caquetá region (present-day Colombia) (Albis 1991). The kind of nurturing relationship that lent itself to hammock sharing could extend to non-humans as well. Everard Im Thurn, an anthropologist in late-nineteenth-century Guyana, mentioned peccaries who 'become very tame—too much so sometimes, for they follow their master wherever he goes and sometimes even insist upon getting into his hammock' (1882, 25–43, 35–36).

11 See also Prudhomme (1798, 353); Roth (1915, 313).

Settler appropriations

European settlers first obtained hammocks in a variety of ways: by conquest, through trade, and as gifts. Colonizing campaigns quickly incorporated the technology, as they sought to dispossess Native peoples of their lands, labor, and, in many respects, their traditions—the very matrix out of which the hammock had originally emerged. These larger military and political processes were premised on a disdain for Native culture and justified by a theory of European cultural and technological superiority. This might suggest that Europeans completely transformed or



Fig. 5: Watercolor of Indigenous courtship rites, with hammocks in background, from *Histoire naturelle des Indes,* fol. 113r. Image credit: Morgan Library, New York, NY

deracinated the hammock, using it in ways that would distance it from its meanings and functions in Native societies. But we suggest instead that, despite European statements disavowing Native technology, settlers adopted hammocks *because of*, not despite, their Indigenous functions and meanings.

Hammocks were sometimes seized as war booty. Hammocks appear, for instance, in inventories of goods confiscated from a Taino *cacique* after his defeat in 1495 and 1496 (Oliver, McEwan, and Casas Gilberga 2008, 228). During the military incursion in Panama led by Balboa in 1514, the Spanish stole cotton hammocks, among other goods (Sauer 1966, 236). Coercion extended to production as well. Colonizers required Native people to produce hammocks within the *encomienda*, the early colonial system that required Native subjects to provide tribute in goods and labor. After Jamaica was invaded and its inhabitants largely subjugated in 1509,

it became a provisioning ground of hammocks. According to Las Casas, 'cotton became the first trade good (*granjería*) that those Spanish in that island had because they forced the people, especially the women, to make great cloths of cotton'—used to make sails—'and shirts, and hammocks that we used for beds and brought to this island [Hispaniola] and Cuba and Tierra Firme' (1875, 3:434; see also Mira Caballos 1997, 244). Archaeological remains indicate that Indigenous laborers produced cotton hammocks for the Spanish as part of the *encomienda* system on sixteenth-century Mona, off the coast of Puerto Rico (Samson et al. 2023). In 1517, the governor of Cuba attempted to force Native people to produce hammocks, but claimed this did not happen 'because in the present on this island there is no cotton from which they can be made' (Mira Caballos 1997, 159, 257). Some *encomenderos* on the mainland likewise demanded hammocks from their tributaries, as was the case in Caricuri, Turbaco, and Guayepo, as well as Cartagena (New Granada/Colombia); in 1549, Indigenous tributaries in the former were required to give the *encomendero* four hammocks annually (Gómez Pérez 1984, 139, 188, 253).

In addition to coercion, trade remained an important way for colonists to acquire hammocks throughout the early colonial period and beyond. In the reports that local officials sent to the Spanish Crown in the late sixteenth century, hammocks appear as items that settlers in Cumaná and Margarita Island sought in barters with Indigenous communities (Latorre 1919, 89). And, in the early seventeenth century, Spanish settlers traded with allied Arawaks in the Orinoco watershed to obtain hammocks, among other goods (Vázquez de Espinosa 1948, 66). French traders likewise acquired hammocks from Kalinago groups in the eastern Caribbean (Tertre 1654, 425). Eder, the Jesuit missionary who lived among the Moxos in the western Amazon, complained that 'some Spaniards' who were only interested in 'trade and profits' bought 'from the ignorant neophytes [Moxos] a shirt, a hammock, and many pounds of cotton' in exchange for 'a few pieces of glass' and 'some worthless bells' (1888, 176). French settlers in Cayenne bought hammocks from Indigenous traders at the mouth of the Amazon in the late eighteenth century (Prudhomme 1798, 331).

However they were acquired, hammocks became an essential part of colonists' military toolkit. In this, they were in part imitating Indigenous combatants, who likewise used them when they went to war (see Léry 1990, 160). Hammocks routinely appear in reports of *entradas*, as military campaigns were known in Spanish. Las Casas, for instance, makes a casual mention of a Spaniard sleeping with a sword in his hammock (1875, 5:13). Oviedo wrote about their suitability for those 'engaged in war' (1535, fol. 80v). Likewise, the commander Nicolas Durand, sieur de Villegaignon, who oversaw the French settlement in what is today Rio de Janeiro,

ordered an enslaved Indigenous man to build 'a little house' in 'native fashion,' and, recalled Léry, 'in the style of the Americans, we hung sheets and cotton beds there, so that we slept suspended in the air' (1990, 35). In 1595, Walter Raleigh, leading an English expedition into Guyana aimed at contesting Spanish control, noted that his men used hammocks and that 'all the Spaniards' in hot countries used hammocks (2006, 85). Missionaries embarking on expeditions customarily brought hammocks along as well, as recorded by Eder in the eighteenth century (Eder 1888, 126–27).

Hammocks were so central to colonists' military goals in the Americas that some envisioned their adaptation as a military technology in Europe as well. Hammocks, Oviedo wrote, 'would not be without small benefit for Spain and Italy and other parts because not so many would suffer or die because of sleeping on the ground in winter and stormy weather' (1535, fol. 80v). Léry, too, thought they belonged in Europe: 'In time of war,' he wrote, referring to his experiences in the War of Religion,

> it is far easier to suspend these sheets in the guardroom for some of the soldiers to sleep on while the others stand guard, than to have them sprawl in the usual way on the straw, where you soil your clothes and get them full of vermin, and find when you get up to be ready for duty that your ribs are bruised from the arms that you must always have at your belt (as we did during the siege of Sancerre, where the enemy did not move from our gates for a whole year). (1990, 160–61).

Walter Raleigh similarly recommended them for European soldiers (2006, 84).

Over time hammocks became not only part of expeditionary equipment but also an important part of settler-colonial material culture more generally (Samson et al. 2023; Zimmerman 2017). By 1519, on the island of Mona, part of the Puerto Rican archipelago. Spanish officials categorized the hammock as an essential item: for instance, it was one of the items—along with bread, maize, beans, and salt—that appeared in a report about the prices of essential goods (Seville, Archivo General de Indias, Patronato 175, R. 6). Hammocks had so fully become part of French settler culture in the Caribbean by the 1650s that Breton could distinguish between Kalinago and French ways of sleeping in hammocks: the French used mattresses and slept lengthwise (*a la Francoise*), while the Kalinago slept across (*travers*) and used the hammock itself to make a cover (1665, 193). Breton's anecdote shows us both the centrality of the technology to French life in the Caribbean and also how Europeans adapted pre-existing sleeping practices to the new technology. Sloane similarly observed that in Jamaica 'hamacas are the common Beds of ordinary white People, they were in use amongst the Indians, and are

much cooler than Beds, so cool as not to be lain in without Clothes, especially if swung' (1707, xxx).

The fact that hammocks had become part of the toolkit of colonial campaigns might suggest that the European adoption of this Indigenous technology was completely divorced from its situatedness in Native psychosocial relationships. Yet, this was not the case. Careful attention to early sources suggests that settler Europeans adopted the technology because their earliest exposures came from their immersion in core rites of care and intersubjectivity, hospitality and healing.

Fleeting traces in the documentary record point to the fact that Native people introduced Europeans to hammocks because of their fundamental importance in friendship and alliance. Within the first week of Columbus' landing in the Caribbean, the Indigenous islanders had given the Europeans cotton hammocks (Las Casas 1875, 1:311; Oliver, McEwan, and Casas Gilberga 2008, 205). In 1496, when Bohechio, the ruler of Xaragua—one of the Taino polities on the island that the Spanish called Hispaniola—still thought that the hairy visitors could become allies, he hosted some of them in lodgings with hammocks, described as 'their beds . . . of cotton, very beautiful, for those who were rich' (Las Casas 1875, 2:139). Juan de Castellanos, a conquistador who lived among Indigenous communities in late sixteenth-century New Granada (Colombia), described the hospitality that Indigenous communities showed Europeans by 'suspending hammocks' (1955, 97). Rochefort, in the eastern Caribbean, similarly noted that, after exchanging greetings, the first step in Kalinago hospitality was 'to present their Guests with those pensile Beds which they call *Amais,* very clean and white, whereof they have store against such occasions: they desire them to rest themselves thereon' (1666, 310). This practice among Tupinamba in Brazil was illustrated in the first edition of Thevet (Fig. 6).

Sick or exhausted Europeans were exposed to hammocks when Indigenous allies cared for them. During Amerigo Vespucci's expedition on Curaçao, he claimed: 'if any of our people got tired on the way, they carried them in their nets very comfortably.'[12] Castellanos, too, mentioned Indigenous hosts not only providing hammocks for 'sleep' but also as 'cures' for the sick among them.[13] Hans Sloane specified that hammocks benefited those suffering from rheumatism or 'inflam'd Blood,' as well as 'Women with Child in danger of Miscarriage' (1707, xxx). It may also be that Europeans acquired the idea of being carried around in hammocks because they interpreted that action as one of deference, when in fact it was of compassion or even pity. Indeed, Sloane made such a connection, rather inadvertently, when he wrote that a hammock is 'a very good way and easie for a Sick Person to be carried on one of these from place to place

12 Vespucci (1894, 15). Vasco Núñez de Balboa was also carried by Indigenous people when sick. See Las Casas (1875, 4:128).

13 Castellanos (1955, 97); Aguado (1906, 26).



Fig. 6: Hospitality ritual taking place in a hammock. From André Thevet, *Les singularitez de la France Antarctique* (Paris, 1588), 85–87. Image credit: Huntington Library, San Marino, CA

by four Men, as is the custom of the better sort of people in Guinea' (1707, xxx).

A few accounts describe in more detail the interactions that led the Europeans to learn about and ultimately embrace this technology. Jean de Léry revealed his appreciation for and slight ambivalence about hammocks at length: 'I ask those who have tried them out whether you don't sleep better on these cotton beds than on our ordinary ones, especially in summer' (1990, 160). He wrote about his initiation into hammocks, about the 'ceremonies that Tupinamba observe when they receive friends who go to visit them,' based on his own experiences 'during [his] first journey among the *sauvage*s' (1990, 164–66). After 'a visitor'—Léry here refers to himself in the third person—was 'on a cotton bed suspended in the air,' he was expected to remain silent for a short period (1990, 164). 'Then the women come and surround the bed, crouching with their buttocks against the ground and with both hands over their eyes . . . weeping their welcome to the visitor, they will say a thousand things in his praise . . . and if it's a Frenchman or some other stranger from over here, they will add "You

have brought us so many fine things that we do not have in this country"' (1990, 164). He explained that 'if the newly arrived guest who is seated in the bed wants in turn to please them, he must assume the appropriate expression, and . . . must heave a few sighs and pretend to weep' (1990, 164). The women conduct this part of the ceremony. The male host (the *moussacat*) will 'meanwhile have spent a quarter of an hour or so pretending not to see you,' occupying 'himself with making an arrow or some other object, a blandishment quite contrary to our embraces, hugs, kisses and handclaps upon the arrival of our friends' (1990, 164). Then the Tupinamba man will ask him his reason for coming and offer food and drink. The women will bring 'him fruit or some small gift from their region, to obtain from him combs, mirrors or the little glass beads that they put around their arms' (1990, 164). The hammock hospitality did not end there:

> if, moreover, you want to sleep in that village, the old man will not only stretch for you a fine white bed, but also, even though it does not get cold in their country, he will place around the bed, against the night's humidity, four or five small fires, which will often be relit in the course of the night, along with some little screens that they call *tatapecoua*, made like masks that the ladies over here hold in front of them when they are next to the fire, to keep it from spoiling their faces. (1990, 164)

In addition to being an invaluable trace of the way that hammocks were an essential technology in customs of hospitality which laid the basis for gift exchange, Léry's reminiscences also reveal in unusual detail the kinds of experiences that led to European adoption of the technology.

If Léry's account illuminates the kinds of interactions that introduced Europeans not only to the material properties of hammocks but to their role in care work, Oviedo's writing shows us how even settlers with the most explicit anxiety about Indigenous cultural practices nonetheless became converts to this technology. In the *Sumario* (1526), his first publication about the Americas, he quickly explained hammock form and function, praising the fine cotton and concluding simply that 'they are very clean and one can sleep quite comfortably in them' (1959, 42). His later *Historia natural de Indias* (1535) expands on this. The hammock appears alongside tobacco in a chapter whose stated purpose was to describe the 'vile vices' of Indigenous people in order to justify brutal colonial tactics. Oviedo first details how 'the Indians consid[er] this particular herb [tobacco] very precious' and 'very sacred' (1535, fol. 47v). When the *cacique* or leader (principal) falls down on ground after using tobacco—likely mixed with a potent hallucinogenic *anadenanthera peregrina*—the women (who are many) take him and lay him in bed or hammock, if he

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Fig. 7: Image of a hammock slung between two trees, with a naked Indigenous figure inside. Gonzalo Fernández de Oviedo y Valdés, *Historia general de las Indias*, Madrid, Real Academia de la Historia, MS 9-551, fol. 130r. Image credit: Real Academia de la Historia, Madrid

told them to do so before he fell' (1535, fol. 47v). Though an intentionally negative caricature of Indigenous practices, the text nonetheless reveals that Oviedo correctly understood the connection between hammocks and shamanic practice, and perhaps sexual intimacy as well. His sketch of a hammock, too, in one early manuscript version, emphasized its connection to indigeneity: the figure reclining in the hammock is depicted as a Native person, as indicated by their nudity (Fig. 7).

And, yet, despite describing the hammock as a quintessentially Indigenous technology, and one implicated in practices that colonizers used to justify colonialism, Oviedo seems to have happily adopted it. He acknowledged the initially uncomfortable kinesthetic experience for those unfamiliar with hammocks. 'In truth,' he wrote, 'for those who are not

accustomed to these beds, they are not that comfortable if they are not very wide; the head and the feet of those who sleep in them are higher than the back, which is low' (1535, fol. 47v). But Oviedo then reveals his own embrace of hammocks—though he maintained some distance by using the third person—when he writes that 'they are good and clean beds' and that they make covers unnecessary, for there is 'spare cloth . . . if they want to fold it over themselves,' or one can build a fire 'under the hammock on the ground or close by to warm themselves' (1535, fol. 47v). Oviedo included another illuminating passage about hammocks in a chapter about the 'large and beautiful' *jobo* trees (hog plum or *Spondias mombin*), valued by Indigenous people throughout the Circum-Caribbean and South America for their delicious yellow fruit, their bark, which could be used for bathing and to relieve 'fatigue after walking,' and their roots, which were a water source. He wrote that when voyaging in the woods and needing to sleep, men 'always look for these *hobos* when they need to sleep in order to suspend their hammocks' (1535, fol. 80v). He explained that travelers 'prefer to be beneath the *hobo*, because its shade protects against nighttime dew and doesn't lead to sorrow or headache as many other trees tend to do' (1535, fol. 80v). Oviedo also conveyed the importance of selecting trees for hammocks by choosing to draw different species of trees in the sketch he made in his manuscript (Fig. 7). Here, Oviedo showed that Europeans like himself, in imitation of Native people, had come to understand that hammocks were not just for sleeping, but also for providing a certain, restful state of mind. He inadvertently revealed the inextricability of mind and body states, albeit in only the faintest outlines, in European adoptions of hammocks. He showed how they continued to be associated with rest and recuperation that went beyond utilitarian notions of sleep.

Conclusion

Oviedo's reactions to the hammock are symptomatic of larger tensions accompanying the settler adoption of Indigenous technologies. Colonizers insisted that European technologies, in aggregate, were superior to Native ones, and this claim of technological superiority figured prominently in their ideological justifications for dispossession (Pagden 1990, 13–36). For instance, one of the most influential apologists for Spanish colonization, Juan Ginés de Sepúlveda, justified the 'sum total of their misfortunes' that would result from Indigenous people 'being forced to change their rulers' because of 'the good things that his war brings about' (2023, 144). These 'good things' were, above all, the superior technologies that European colonization would bring: domesticated animals, writing, and iron tools.

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Sepúlveda explained that 'the kings of Spain are, as it were, the inventors of so many very useful and necessary things unheard of in those regions,' asking 'with what acts of reverence, what duties, and what honours could the barbarians render thanks equal to so many and such great benefactions?' (2023, 145).

Yet, the European claim for technological superiority over Native groups is belied by the fact that the colonizers coveted many Indigenous technologies. Settler reactions to hammocks exemplify the central predicament of Indigenous technologies under colonialism: the gap between what apologists for conquest and dispossession *said* about Native technologies, and what they actually *did* with them. As is also true with another, related technology, tobacco, Europeans' initial adoption of the hammock was because of, not despite, meanings invested in and practices surrounding this Indigenous technology (Norton 2008). The process by which European settlers came to adopt this technology cannot be separated from the hammocks' entangled material and symbolic qualities in their Indigenous contexts throughout the Caribbean and South America.

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